AI, JUDGMENT AND THE FUTURE OF FINANCE

Four Essays for Financial Practitioners

Institutions, agency, the exo-brain, and moral responsibility

ALEJANDRO REYNOSO

LECTURE 1 From Models to Autonomous Institutions

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MICHAELMAS 2026 | INTRODUCTORY NOTE

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Four Essays for Financial Practitioners — Michaelmas 2026

This collection brings together four essays accompanying the sequence of three lectures of *AI for Financial Practitioners*, offered during Michaelmas 2026 at Cambridge Judge Business School, University of Cambridge.

Each essay explores a different dimension of our exploration of artificial intelligence. The first, *From Models to Autonomous Institutions*, introduces the transformation of AI from a technological capability into a matter of institutional leadership. The second, *Who Is Artificial Intelligence?*, invites us to reconsider AI not merely as a tool, but as an increasingly active participant in organizational processes, raising fundamental questions about authority, judgment, and responsibility. The third, *The Exo-Brain*, explores how AI can extend our capacity to understand, create, and accomplish things through new forms of cognitive and operational collaboration. The fourth, *The Architecture of Reason*, turns to Dante and the classical architecture of moral responsibility to examine deception, provenance, and the conditions of institutional trust.

These are not technical lectures or comprehensive treatments of their subjects. They are invitations to think. Their purpose is to establish a common intellectual starting point, encourage curiosity, and connect emerging AI capabilities with the professional experience and judgment that financial practitioners bring to the classroom.

Together, they reflect a simple conviction: understanding AI is becoming an essential part of understanding the future of finance, its institutions, and our responsibilities within them.

Alejandro Reynoso
Cambridge Judge Business School
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OPENING ADDRESS MICHAELMAS 2026

FROM MODELS TO AUTONOMOUS INSTITUTIONS

The new responsibility of financial leadership

From the models we know to the institutions we must learn to govern: an opening address for the Michaelmas course in artificial intelligence for financial practitioners.

ALEJANDRO REYNOSO OCTOBER 2026

Ladies and gentlemen, colleagues,

Imagine that two investment memoranda arrive on your desk tomorrow morning. They concern the same company. They describe the same deterioration in margins, the same increase in debt, and the same uncertainty about refinancing. Both are lucid. Both are persuasive. One required several days of professional work. The other appeared in a few minutes.

That difference deserves our attention. But before we decide what it means, we should ask a less spectacular question. Which memorandum gives us a sounder basis for a decision?

The answer is not visible in the elegance of the prose. We need to know whether the figures refer to comparable periods, whether the refinancing belongs to the same legal entity, whether an assumption has quietly become a fact, and whether the recommendation respects the constraints of the portfolio. We need to know what was checked, what remains uncertain, and what would happen if we acted.

This is where I would like to begin our conversation about artificial intelligence: at the distance between an impressive answer and a dependable institutional capability.

For financial leaders, that distance is becoming one of the most consequential spaces in management. Within it lie questions about productivity and investment, certainly, but also questions about knowledge, organization, authority, and responsibility. The technology invites us to reconsider how an institution thinks and acts. Our task is to make that reconsideration precise enough to guide decisions.

I would like to take you on a journey from the models we have long understood to the systems we are beginning to build, and then toward the more adaptive organizations we can now investigate. The destination is an institution able to combine capable machines with professional judgment, sound evidence, and a clear account of who is responsible for what.

Finance has always been a discipline of representations. A balance sheet represents an enterprise. A valuation represents a set of expectations. A risk model represents selected relationships between an uncertain world and commitments made today. We have never held the future itself in our hands. We have worked with ways of describing it.

Statistical models brought a particular discipline to this work. They asked us to specify relationships, expose assumptions, estimate parameters, and consider uncertainty. Their value included the opportunity to disagree intelligently. We could ask why a variable entered an equation, whether the sample was appropriate, or whether a relationship reflected causation, selection, or coincidence.

Machine learning expanded the repertoire. More flexible methods could identify patterns that restrictive specifications might miss. This enlarged what we could attempt, while making evaluation beyond the observed sample even more important. A model that explains yesterday beautifully can still disappoint tomorrow. In finance, the temptation to allow tomorrow's information into yesterday's test is especially dangerous, because the resulting success can look remarkably convincing.

Representation learning expanded the kinds of material that could enter the analysis. Language, images, sequences, and networks could become inputs to useful computational comparisons. Information that had previously required extensive manual interpretation became more accessible to systematic treatment.

Then generative models brought a striking change in our experience of the technology. We could ask a question in ordinary language and receive an explanation, a draft, a proposed calculation, or working code. Architectures such as the Transformer helped make contextual relationships in language computationally tractable at scale. The practical interface became a conversation.

That conversational quality matters. It lowers the cost of exploration. A finance professional can investigate an unfamiliar method, compare alternative interpretations, or prototype an analytical idea without first mastering every implementation detail. This creates room for initiative across the organization. Ideas that might once have remained unexpressed can become something colleagues can inspect and improve.

Yet the apparent unity of the interface can conceal the diversity of the work. Inferring a relationship, predicting an outcome, interpreting a document, generating an explanation, and choosing an action are different accomplishments. Each calls for a different kind of evidence. A fluent explanation cannot, by itself, establish that a forecast is calibrated or that a transaction is appropriate.

Earlier methods therefore remain part of the architecture. A language model can help interpret a disclosure, a statistical model can estimate a probability, a calculator can perform exact arithmetic, and an optimization routine can evaluate a constrained choice. The financial professional gains a larger repertoire and a more demanding responsibility to compose it well.

Even apparently simple words deserve care. Training changes a model's parameters. Context supplies information for a particular invocation. Retrieval selects external evidence. Memory preserves information across interactions. Uploading a report does not necessarily retrain a model, and remembering a statement does not establish its truth. These distinctions help an executive understand what a proposed system actually does.

Consider a margin calculation. The arithmetic may be

elementary. The interpretation may depend on whether revenue is recurring, costs are comparable, and exceptional items have been treated consistently. We can delegate the arithmetic with great precision while retaining an unresolved economic question. Good architecture makes that distinction visible.

Our first intellectual transition is therefore from asking which model is most impressive to asking which capabilities the work requires. That question brings us back to the institution's purpose. It also prepares us for the next transition, because a capability must operate through time if it is to become a service.

Let us return to the memorandum. Now imagine asking the institution to produce it continuously for five hundred companies. New filings arrive at different times. Some contradict earlier disclosures. Some describe subsidiaries with similar names. Reviewers are unavailable. Data feeds fail. A recommendation prepared at the wrong moment may lose much of its value.

The assignment has changed. We need a process that perceives, judges, executes, and learns from consequences. These four functions give us a useful way to understand an intelligent system. Governance surrounds the whole cycle, because the permission to obtain information, interpret it, and act upon it must be considered at every stage.

Perception begins with constructing a usable account of the situation. The system must know which entity a statement concerns, which period it describes, where it came from, and when it became available. Missing information must remain visibly missing. An unknown figure quietly replaced with a plausible estimate can contaminate everything that follows.

Suppose a fictional company, Northbridge, reports that debt falling due within twelve months has risen sharply. A later announcement describes refinancing. The words appear reassuring. But does that announcement concern the same subsidiary and the same obligation? Until the relationship is established, the reassuring interpretation remains a hypothesis. This is a question about the identity and meaning of evidence before it becomes a question about financial judgment.

Judgment then asks what the evidence implies in relation to an objective. A deterioration in liquidity may justify further investigation, a revised valuation, or a proposed change in exposure. The choice depends on uncertainty, alternatives, constraints, and the consequences of error. Prediction contributes to this decision, but it cannot determine the institution's objectives for it.

Planning connects that judgment to an executable course of action. Which steps are required? In what order? Which dependencies must be resolved? Optimization may help identify a preferred choice within a permitted set, but a mathematically attractive choice outside that set has no institutional authority.

Execution introduces another distinction. Saying that something has happened is different from confirming that it happened. Imagine an internal alert sent to a review queue. The acknowledgment fails to arrive. An immediate retry might be sensible, or it might create a duplicate action. The service must establish what the receiving system actually accepted. Operational intelligence includes the ability to handle uncertainty about its own effects.

Feedback closes the cycle, but only if we understand what the feedback tells us. A reviewer liking the prose is evidence about usability. A source supporting a claim is evidence about its grounding. A forecast matching subsequent outcomes contributes evidence about predictive performance. These observations belong to different questions. Praise is welcome; it is an inadequate substitute for attribution.

This is why a successful demonstration is only the beginning of evaluation. We must examine ordinary cases, difficult cases, and cases at the boundary of the system's authority. We must ask how it behaves when evidence conflicts, when an operation is interrupted, and when the responsible response is to stop.

The quality of the service lives partly in the handoffs. A number can be correct and still lose its meaning when its units disappear. A conclusion can be reasonable and still become dangerous when its conditions are omitted. We are designing continuity of meaning across a sequence of operations. That is a substantial management accomplishment.

As the service grows, it becomes useful to distribute its responsibilities. We begin to speak about tools, skills, memory, agents, and teams of agents. These terms can sound like a catalogue of products. Their deeper value is organizational: they allow us to say which part of the work belongs where.

A tool performs an operation. It searches a document, calculates a quantity, or writes a record. A skill, in the vocabulary I am using, is a reusable procedure: how to review a covenant, reconcile a disclosure, or prepare a sourced memorandum. It may employ several tools and should specify what counts as satisfactory completion.

Memory gives the process continuity. It allows the next assignment to benefit from what happened before. But a financial institution needs more than persistence. It needs distinctions between a tentative observation, a proposed interpretation, and an accepted conclusion. Otherwise, a mistake can acquire the appearance of authority simply by being retrieved repeatedly.

This takes us into three subjects that may sound philosophical but are intensely practical: semantics, ontology, and epistemology. Semantics asks what a statement means. Ontology asks what kinds of things exist in our working description of the business and how they relate. Epistemology asks why we believe a claim and what evidence could lead us to revise it.

An obligation belongs to an entity. A guarantee connects parties under specified conditions. A reporting period has a beginning and an end. If our system confuses these relationships, more eloquent language will not repair its understanding. Likewise, a claim stored in an elegant knowledge graph remains a claim. Its representation cannot confer truth upon it.

The question becomes: what does the institution permit itself to know? Who can promote a hypothesis into the accepted record? How is a correction propagated to the conclusions that depended on the earlier claim? These are

familiar responsibilities in research, accounting, and risk management. AI makes their design more explicit because knowledge can now be reused at considerable speed and scale.

An agent adds discretion over a sequence of operations. Given a bounded objective, it may decide to retrieve another document, request a calculation, or revise an interpretation. A fixed workflow specifies the path more closely in advance. Both can be valuable. We should allocate discretion where variation in the task makes it useful.

An agent also needs stopping conditions. It may have completed the assignment, exhausted its budget, encountered inadequate evidence, or reached a decision requiring escalation. Its declaration of success should be tested against the assignment's acceptance criteria. Otherwise, we have delegated both the work and the right to pronounce it satisfactory to the same unexamined process.

A team can separate evidence gathering, calculation, synthesis, and review. This creates opportunities for specialization and challenge. It also creates dependencies, communication costs, and disagreement. Coordination is real work. Adding agents increases the number of relationships we must understand as well as the number of tasks we can perform.

Imagine two reviewing agents that agree enthusiastically. Their agreement is reassuring only to the extent that their methods can uncover relevant errors. If both rely on the same outdated summary, their consensus adds little. A reviewer that returns to the original evidence or recomputes a material quantity may provide a much stronger check.

We have reached a revealing point. Once intelligence is distributed, questions about its effectiveness begin to resemble questions about organizations: who specializes, who challenges, who integrates, and how disagreement reaches the decision maker. The executive's existing experience becomes highly relevant. It must, however, be translated into observable procedures rather than human job titles assigned to software.

The next step is consequential. The service works well enough that someone asks to let it update approved records or initiate actions without waiting for a person

each time. At that moment, the institution is making a delegation decision. Even if the underlying model is unchanged, the system's relationship with the world has expanded.

Autonomy is best understood through the discretion granted for a particular purpose. A service may choose which approved documents to read while having no authority to contact a client. It may update routine status fields while requiring approval to alter an investment conclusion. Its discretion can be limited by time, budget, information scope, and action class.

We can think of a progression from advice, to individually approved action, to bounded execution, and then to controlled adaptation. These are useful distinctions for management. They do not represent a competition in which the institution with the greatest autonomy wins. A highly capable service may properly remain advisory because the decision requires responsibility that the institution retains elsewhere.

At the same time, the object of governance expands. We begin by examining a model. We then govern its tools and workflows, its institutional knowledge, its continuing activity, its security exposure, its collaboration with other agents, and eventually the behavior of a more complex interacting system. Earlier obligations accumulate as the unit becomes larger.

This means we cannot postpone cybersecurity until the architecture becomes advanced. Nor can we stop caring about source quality once agents begin to collaborate. A sound component contributes to a sound system, but it does not establish one. Dependencies and interactions create properties that component reviews alone may miss.

Policies express our intentions. Controls make those intentions operational. If the policy prohibits external messages, the available credentials and release mechanisms must reflect that prohibition. A written instruction asking an agent to behave is one element of the design. It needs support from mechanisms that restrict, detect, and correct consequential actions.

Authority must also remain intelligible when work is delegated again. A specialist should receive the access needed for its assignment. The coordinator's broader access should not travel automatically with every message. We need to distinguish who initiated the work, who selected an action, what identity executed it, and who accepted the result.

Human oversight deserves equally concrete treatment. A person in the loop needs evidence, time, competence, and a feasible workload. If a reviewer receives hundreds of polished proposals and seconds to approve each one, the architecture may produce the appearance of oversight while depriving it of substance. Management must design the conditions in which a human can make a meaningful difference.

We should be equally precise about stopping. Closing an interface may not halt jobs already running elsewhere. Revoking a credential may not reverse an action already accepted. Recovery begins by establishing what completed, what remains pending, and what is uncertain. Some records can be restored; some consequences require correction or remediation.

There is a deeper leadership lesson here. Delegation can be widened, narrowed, or withdrawn as evidence changes. A well-designed boundary gives us the confidence to experiment because we understand the conditions under which the experiment can proceed. Authority becomes a deliberate, revisable institutional arrangement.

Now imagine that the environment changes. Liquidity becomes more important. The existing research process generates more exceptions. A useful question arises: could the system recognize the change and reorganize some of its work?

We must first distinguish response from adaptation. A fixed rule can reduce exposure when volatility rises. The action changes because the observation changes. The rule remains the same. Adaptation changes something in the mechanism itself: perhaps a parameter, a routing procedure,

a memory instruction, or the choice of specialist.

This leads to two connected loops. The inner loop performs the task. It asks what the current filing means and what action is permitted. The outer loop examines the way the task is performed. It asks whether repeated failures justify changing the process. The two loops need different evidence and may require different authority.

A supervisor might activate a liquidity specialist from an approved menu. That is an organizational change we can describe and test. We can compare it with the fixed arrangement, observe the extra cost and delay, and ask whether the new specialist resolves the failures that motivated the change. We can also discover that waiting for more evidence is preferable to reacting immediately.

Learning can occur around a model without changing its underlying parameters. A system may retain a lesson from an earlier attempt or adopt a revised procedure. But its explanation of a failure may itself be wrong. A plausible retrospective story becomes useful learning only when it improves subsequent work under a meaningful test.

Recursive improvement takes us one step further. The system proposes changes to the procedure that proposes improvements. We are now examining the improver as well as the activity it improves. Bounded experiments in this direction are intellectually important, but they should not be confused with evidence of unrestricted, reliable self-improvement.

The central danger is circular evaluation. If a candidate system can redefine the objective, remove difficult cases, and decide whether its own change succeeds,

apparent progress becomes very easy to manufacture. The institution must preserve an independent account of what improvement means. Protected requirements can change through governance; they cannot simply dissolve inside the search for a better score.

Beyond assigned teams lies the possibility of endogenous organization. Agents may select partners, develop specializations, or retain successful working relationships. Some of the division of labor then emerges through interaction. The term swarm is useful here when it directs attention to collective behavior whose detailed organization was not fully prescribed in advance.

At this point, our inquiry begins to resemble computational sociology. How does knowledge circulate? Which participants gain influence? Why do coalitions form? Do incentives reward accuracy, speed, or persuasion? Useful specialization may emerge, but so may concentrated dependence or the exclusion of inconvenient evidence.

Game theory gives us questions to ask about these interactions. We can investigate whether cooperation benefits the participants and whether a participant could gain by deviating. But a convincing conversation among agents does not establish an equilibrium. If we test only a finite set of deviations, our conclusion must remain limited to what those tests actually examined.

Ecologies extend the question to persistent systems with different owners and purposes interacting across boundaries. Useful information may cross from one system to another. Authority need not follow it. A productive relationship with an outside specialist does not automatically justify access to confidential records or consequential tools.

These possibilities belong in disciplined experiments. Compare a fixed organization with an adaptive one, but also ask whether keeping the extra specialists permanently active explains the same benefit. Account for resources. Preserve failed attempts. Hold back scenarios from the development process. A simulation can reveal a mechanism and its limitations; evidence for production requires further work.

The future becomes more interesting when we admit these distinctions. We can study systems that change their organization without pretending that every change is progress. We can investigate emergence while retaining responsibility for the environment that makes it possible. Curiosity and rigor can reinforce one another.

This brings us to the strategic position of the financial institution. It must protect commitments already made while developing capabilities that may transform how future commitments are fulfilled. I describe these responsibilities through two images: the Fortress and the AI Factory.

The Fortress contains what must remain dependable: client information, authoritative records, critical services, permissions, continuity, and recovery. Its purpose is to preserve the conditions under which the institution can honor its obligations. As AI connects more sources, tools, and actors, those conditions require renewed attention.

A malicious instruction can arrive disguised as material to be analyzed. An overly broad credential can turn a narrow analytical error into a wider operational event. A compromised memory can influence many later decisions. We therefore need to understand what an attacker could cause the whole system to do, alongside the quality of its legitimate outputs.

The AI Factory creates room for productive exploration. It tests ideas, develops reusable capabilities, records failures, and evaluates candidate services. Synthetic data and appropriately authorized historical material can support much of the early work. Clear boundaries give talented people room to experiment with confidence.

The connection between these two domains is the heart of the strategy. An experiment produces evidence. Shadow operation tests behavior alongside the existing process. A bounded pilot grants limited authority. Maintained service adds continuing ownership, monitoring, and funding. Each transition is a decision about evidence, investment, and responsibility.

Financial discipline must follow the entire service. The price of a model call is only one element of cost. Data preparation, retrieval, integration, review, rework, security, monitoring, and maintenance all contribute. An inexpensive answer can become an expensive process if it creates repeated exceptions or shifts repair work onto professionals whose time is not being measured.

Consider a purely illustrative monthly comparison. The existing process costs thirty thousand units and produces nine thousand accepted results. A proposed service costs twenty-five thousand units for the same number, under the same acceptance standard. That implies a saving of five

thousand units. The claim becomes meaningful only when the totals include the work needed to make the results useful.

Redeployed capacity deserves its own account. If analysts spend less time reconciling documents and more time examining difficult decisions, that may be valuable even when payroll does not fall. Faster information, broader coverage, and better records may also matter. We should state these benefits clearly and evaluate them through evidence appropriate to each claim.

Ownership completes the design. Someone in the business must own the purpose and accept operating outcomes. Technology and operations must maintain the service. Data, risk, and security functions must shape its boundaries and challenge

the evidence. The board must understand the consequential delegations and the basis for continuing or expanding them.

A pilot that continues indefinitely without an explicit decision has not necessarily become an institutional capability. It may have become an unattended dependency. Maturity requires the ability to explain why the service exists, what it costs, how it is performing, and what would lead us to change or retire it.

Let us return, finally, to the two memoranda on your desk. We can now see how much lies behind the choice between them. We have traveled from the quality of a model to the integrity of a process, from the division of work to the allocation of authority, and from adaptation to the strategy of the institution itself.

The executive opportunity is substantial. More people can explore ideas. More evidence can enter a decision. Routine work can become faster, and professional attention can move toward ambiguity, judgment, and relationships. Realizing that opportunity requires us to make the surrounding organization worthy of the capability we place inside it.

There is also a question about how expertise is renewed. Much professional judgment develops through work that initially appears routine: reconciling a number, tracing a discrepancy, discovering why an apparently simple comparison fails. If we automate those encounters, we should consciously create other ways for younger colleagues to acquire the understanding they once provided. Productivity and professional formation belong in the same design. An institution needs people able to question its systems, including after the people who built them have moved on.

For experienced practitioners, the change can be equally demanding and liberating. We may spend less time assembling a first account of a situation and more time testing whether it deserves belief. We can ask for alternative explanations, inspect counterexamples, and explore consequences that time previously prevented us from examining. The opportunity is to deepen judgment through a wider field of inquiry. That requires a willingness to challenge an answer even when it confirms our preferred view with unusual eloquence.

We should also consider what happens when similar capabilities become widely available. If many institutions can obtain a persuasive draft quickly, the draft alone becomes a weaker source of distinction. The quality of the institution's knowledge, its relationships, its evaluation practices, and its ability to coordinate action may become more consequential. Competitive advantage can arise in how these elements work together. That is something leadership can cultivate over time, through choices that no procurement contract can make on its behalf.

Perhaps the most valuable habit we can develop is the ability to hold two attitudes together: intellectual openness about what these systems might make possible,

and exactness about what the evidence presently supports. Openness gives us the courage to investigate. Exactness helps us recognize whether the investigation has succeeded. Together they make it possible for an institution to learn without needing every experiment to vindicate the original enthusiasm of its sponsors.

Our professional identity will evolve with that task. Financial expertise remains essential, because objectives, materiality, uncertainty, and consequences need interpretation. What grows around that expertise is an architectural responsibility: the ability to connect intelligence with evidence, evidence with action, and action with accountable ownership.

We need not predict a single inevitable destination. Different institutions will delegate different things, at different speeds, for different reasons. Their choices should be intelligible in terms of the work they undertake and the commitments they carry. The possibility of greater autonomy gives us more decisions to make; it does not make those decisions for us.

The questions I would leave with every leadership team are these. What are we trying to accomplish? What evidence would justify trusting this arrangement within a defined boundary? And what would cause us to intervene, revise it, or withdraw its authority?

Those questions connect our past to our future. They preserve the financial discipline of examining assumptions while opening the institution to new forms of intelligence and organization. They allow us to be ambitious with a clear understanding of what our ambition requires.

We began by learning to build models of the world. We are now learning to build systems that participate in it. Ahead of us lies the work of shaping institutions that can learn, adapt, and act while remaining answerable for the consequences. That is a demanding responsibility. It is also an extraordinary opportunity for financial leadership.

Thank you.

MANUSCRIPT NOTE

This continuous address is adapted from Alejandro Reynoso's AI for Financial Practitioners: From Models to Autonomous Institutions, complete chapter draft 0.2, 18 September 2026. It covers the six chapters and their final institutional synthesis. The source handbook contains the supporting references.

Northbridge and the cost comparison are hypothetical teaching examples. References to adaptation, swarms, and ecologies distinguish conceptual possibilities and bounded experiments from established production capability.

The spoken text contains approximately 4,388 words.

Allow approximately 35 minutes at an average of 125 words per minute, including natural pauses. Actual delivery time should be confirmed in rehearsal. This note and the title page are not intended to be read aloud.

The LaTeX source is self-contained and compiles with pdfLaTeX. Comments mark the narrative movements for editing without interrupting the spoken text with section headings.

ESSAY MICHAELMAS 2026

WHO IS ARTIFICIAL INTELLIGENCE?

Why changing a single word may change how we govern the AI age

As artificial intelligence moves from answering questions to participating in institutional decisions, we must ask not only what the technology is, but what place it occupies in the architecture of authority and responsibility.

ALEJANDRO REYNOSO OCTOBER 2026

Welcome back to the classroom. As we begin our second lecture of Michaelmas, I would like to invite you to consider a question that sounds almost grammatical, yet has far-reaching implications for the institutions in which many of you will lead. Our first task is not to abandon the technical understanding of artificial intelligence. It is to complement it with an institutional understanding: how AI participates in decisions, what authority it receives, and who remains answerable for its consequences.

For several years now, in boardrooms, universities, government departments and financial institutions throughout the world, we have been asking what appears to be an entirely sensible question: What is artificial intelligence?

It is a question to which we have devoted an extraordinary amount of attention, and understandably so. We have sought to understand the mathematics upon which artificial intelligence rests, the statistical foundations from which machine learning emerged, the architecture of neural networks, the extraordinary development of large language models, and, more recently, the appearance of agents capable not merely of producing an answer but of using tools, consulting information, retaining memory and undertaking sequences of actions in pursuit of an objective. We have learned a great deal by asking this question, and we should certainly continue to ask it.

Yet I have begun to wonder whether, for those of us whose principal responsibility is not to build these systems but to govern the institutions into which they are entering, we have allowed one particular question to dominate the conversation for rather too long.

Perhaps the most important question is not What is it? Perhaps it is Who is it?

I use the word who deliberately, but not literally. I am not suggesting that artificial intelligence is a person, still less that it possesses consciousness, moral standing or some mysterious inner life merely because it can converse with us. That would replace one conceptual confusion with another. I am suggesting something more modest, but also more consequential: that by changing a single word in our question, we change the frame of reference through which we examine the problem.

Changes of reference can be remarkably powerful. Intellectual history is filled with occasions on which a problem that appeared insoluble within one frame became comprehensible when viewed from another. The mathematics did not necessarily disappear, nor did the physical world change; what changed was the conceptual position from which we looked at it. A new coordinate system, a new mathematical object, a different conception of what should be regarded as fixed and what should be regarded as variable could suddenly reveal relationships that had previously remained hidden. I believe something similar may now be required in our thinking about artificial intelligence.

Changing the question

When we ask what artificial intelligence is, our minds move quite naturally toward its internal construction. We ask how the model was trained, what data it uses, how inference takes place, how many parameters it contains, what computational infrastructure supports it and what technical limitations constrain its performance. Those are legitimate and necessary questions, but they have an institutional consequence that is rarely acknowledged.

Because the questions sound technological, we quite properly summon the technologists to answer them. The engineers arrive, together with the computer scientists, the data specialists and the technology officers, and before the conversation has properly begun we have already made an enormously important decision: we have implicitly classified artificial intelligence as a technological matter.

Now let us change only one word. Let us ask instead: Who is artificial intelligence? Something rather remarkable happens. We cease, at least for a moment, to look inside the machine, and begin instead to look around it.

If artificial intelligence is now participating in the affairs of my institution, who brought it into the institution? Who selected it, and according to what criteria? Who decided what information it should be permitted to know? Who determined which systems it may enter and which must remain beyond its reach? To whom does it report? Who supervises it? With whom is it permitted to communicate? May it communicate with other artificial agents? May it ask them to perform tasks on its behalf? May it use tools? May it initiate an action rather than merely recommend one? If it records feedback from the consequences of its actions, who determines whether that learning should be permitted to alter its future behaviour? Who evaluates its performance? Who may overrule it? Who can stop it? And when an action initiated or recommended by it causes harm, who, in the end, is responsible?

Notice what has happened. We have not changed the model. We have not rewritten a single line of code. We have not altered the database, replaced the servers or modified the mathematics. We have changed only the grammar of the question, and yet we find ourselves discussing an entirely different class of problems. We are now speaking about authority and delegation, supervision and accountability, rights of access and limits of action, institutional responsi-

bility and ultimately governance. We have moved, almost without noticing it, from computer science into organisational theory, from engineering into law, from algorithms into ethics and from technological architecture into the architecture of power.

From systems to the systemic

For those of us who come from finance, this distinction should not be entirely unfamiliar, because we have encountered it before. Consider the expression payment system. It may describe a technological arrangement through which messages are transmitted, transactions are processed, balances are updated and instructions are executed. Such a system may be enormously sophisticated, and its technological resilience may be a matter of national importance.

But now consider the rather broader expression the payments system of a country. We are no longer describing a piece of technology. We are describing an institutional order. Within that order there are commercial banks and central banks, securities firms and clearing houses, central securities depositories and payment processors, corporations and households, regulators and markets. There are contractual relationships, credit exposures, flows of information, flows of liquidity, concentrations of risk and mechanisms for resolving failure.

Understanding the software is necessary, but it is plainly insufficient. To understand the payments system we must understand who participates, who is connected to whom, what each participant may do, what obligations bind them together, where authority resides, how shocks propagate through the network and what happens when one participant ceases to perform its expected role.

At that point, we are no longer studying merely a system. We are studying something systemic. And I suspect that artificial intelligence is now crossing precisely this boundary.

When the tool becomes an actor

For most of the history of computing, our conceptual categories were comparatively comfortable. Human beings acted; machines assisted them. A spreadsheet could be extraordinarily useful, but nobody wondered to whom the spreadsheet reported. A database might contain information of immense value, but nobody asked whether the database should be permitted to negotiate with another database. An optimisation algorithm could solve problems of remarkable complexity, but we did not ordinarily ask whether it should be allowed to assemble a team of other algorithms, distribute tasks among them, evaluate their performance and alter the composition of the team when circumstances changed.

The distinction between actor and instrument was therefore reasonably clear. Artificial intelligence is beginning to make that distinction less comfortable. An advanced AI system can, within defined permissions, be given not merely an instruction but an objective. It can gather information relevant to that objective, interpret what it finds, select among tools, formulate alternatives, examine its own work, consult other systems, retain information from previous interactions and, where it has been given the necessary authority, initiate actions whose consequences extend beyond the computer in which the model resides.

More sophisticated architectures may permit several artificial agents to divide a problem among themselves, specialise, review one another's work, challenge conclusions, escalate exceptions and reorganise their collaboration as conditions change. We need not exaggerate any of this. We need not pretend that these systems are human beings, and we certainly need not attribute consciousness to them. Indeed, anthropomorphism would obscure rather than clarify the problem.

An entity does not need to be a person in order to become an actor within an institutional system.

Financial markets have taught us this lesson repeatedly. Corporations are actors. Central banks are actors. Investment funds are actors. Clearing houses are actors. Committees are actors. We routinely analyse the behaviour of entities whose actions cannot be reduced to the psychology of a single human individual. What matters for institutional analysis is not whether an entity has a human face, but whether it occupies a consequential position within a network of information, authority, incentives and action. That is the sense in which the word who becomes useful.

The hybrid institution

Now imagine a financial institution perhaps in the years ahead. It employs ten thousand human beings, but alongside them operate thousands of artificial agents. Some continuously examine markets; others analyse credit exposures or search for anomalies in transactions. Some prepare investment research, others review contracts, others monitor regulatory obligations, and still others construct scenarios against which portfolios or balance sheets are tested. Certain agents may review the work of other agents, while others may assemble temporary groups of specialists to address a particular problem and dissolve those groups when the problem has been resolved.

At that point, asking how many people work for the institution remains a perfectly meaningful question for the human-resources department, but it is no longer sufficient for understanding how the institution actually functions. The more revealing question becomes: How many actors participate in the production of its decisions?

Once we ask that question, our familiar organisation chart begins to look strangely incomplete.

The traditional organisation chart was designed for a world in which lines of authority connected human beings. It told us that one person reported to another, that departments belonged within divisions and that divisions ultimately reported upward through an executive hierarchy. But the institution I have just described cannot be represented adequately in that manner. What we require instead is something closer to an institutional graph, or perhaps more precisely a map of agency: a representation not merely of people and departments but of every consequential participant in the process of perception, judgment, decision and execution.

Such a map would show human beings, artificial agents, models, databases, tools, committees and external systems. It would show which entities may exchange information, which may issue instructions, which may recommend an action, which may authorise it, which may execute it and which may interrupt it. It would show where human judgment remains mandatory, where authority has been delegated, where several artificial systems interact and, above all, where ultimate responsibility returns to a human institution.

AI enters the boardroom

This is the point at which artificial intelligence arrives unmistakably in the boardroom. A board of directors does not need to become a department of computer science. A director need not learn to construct a transformer any more than the director of an airline must learn to manufacture a jet engine. But directors must understand the architecture through which authority is exercised within the institution for which they bear responsibility. They must know where judgment has been delegated, where risks accumulate, where controls may fail and where accountability ultimately resides.

The board's most important questions about artificial intelligence may therefore prove not to be questions about artificial intelligence at all, at least not in the conventional technological sense. The board must ask where the institution has delegated judgment to an artificial system; whether that system merely advises or may also decide; whether it may merely decide or may also execute; what information it may obtain; what tools it may employ; which other systems it may contact; under what circumstances human approval becomes mandatory; how an unexpected behaviour is detected; how a decision can be reconstructed after the event; and how authority can be withdrawn when circumstances require it.

These are questions that technology specialists must help us answer, but they are not questions that technology specialists can answer for us, because the decision about how authority should be distributed within an institution belongs ultimately to those charged with governing that institution. From ontology to governance And here the argument becomes deeper still. Every institution possesses an ontology, although very few boards of directors would describe it in those terms. An institution necessarily has some conception of the things that exist within its world. There are customers and employees, accounts and transactions, assets and liabilities, committees and departments, authorisations and prohibitions, obligations and exceptions. The information systems of the institution encode these categories because the institution itself could not function without them.

For a very long time, however, one assumption remained largely invisible within that ontology: the entities capable of exercising consequential judgment were ultimately human. Artificial intelligence introduces a new category. It is not necessarily a person, nor an employee, nor a legal subject in the ordinary meaning of those terms. But neither is it adequately described, in every circumstance, as a passive computational instrument. It may occupy a position within the institutional network from which it receives information, forms an assessment, communicates that assessment, uses delegated authority and causes something to happen.

The moment we recognise this, ontology leads naturally to epistemology. If this new actor participates in our decisions, then what does it know? How does it know what it claims to know? Which sources does it regard as authoritative? How does it distinguish evidence from inference? How does it represent uncertainty? Can it recognise the boundary between what it knows and what it merely supposes? Under what circumstances should it seek additional information, refer a matter to a human being or decline to act altogether?

Once we ask how an artificial actor knows, we cannot avoid asking how it ought to act. Epistemology leads us to ethics, because a system capable of participating in consequential decisions must operate within limits that cannot be derived from technical capability alone. The fact that something can be done has never, in a civilised institution, been regarded as sufficient reason for doing it. Ethics, in turn, returns us to governance, because principles without responsibility are merely aspirations. Someone must determine the boundaries. Someone must monitor them. Someone must decide what happens when they are crossed. Someone must remain answerable.

Beginning with a question about a machine, we arrive at the oldest problem of institutional life: the responsible exercise of power.

A different kind of AI literacy

This, I believe, is why our understanding of artificial intelligence must now mature beyond technological literacy. Executives should certainly understand enough about models, data and algorithms not to regard them as magic. Ignorance disguised as delegation is not governance. But neither should we imagine that the executive of the artificial-intelligence age must become a software engineer.

The distinctive responsibility of leadership has always lain elsewhere: in understanding the institution as a whole, in determining what may be delegated and what must remain reserved, in establishing limits to authority, in recognising concentrations of risk, in designing mechanisms of accountability and, perhaps above all, in asking questions that reveal problems before those problems become crises.

The challenge before us is therefore not simply to teach financiers more computer science. It is to develop a generation of leaders capable of crossing the intellectual boundary between finance and artificial intelligence without abandoning the institutional knowledge that finance has spent centuries accumulating. Indeed, that knowledge may now prove extraordinarily valuable.

We already know how to think about networks, principal-agent problems, delegated authority, information asymmetry, moral hazard, systemic risk, contagion, concentration and governance. We know that individually rational actions can sometimes produce collectively disastrous outcomes. We know that systems can display properties that cannot be understood by examining their components separately. We know that trust requires verification, that authority requires accountability and that resilience depends not merely upon

preventing failure but upon designing institutions capable of surviving it. These are not peripheral ideas in the age of artificial intelligence. They may become central ones.

The question for the board

And so I would propose a small experiment for the next board meeting at which artificial intelligence appears on the agenda. When the presentation begins, and the first slide announces the institution's new AI strategy, resist for a moment the temptation to ask which model has been selected, how much it costs, how many parameters it contains or which technology company provides it.

Who is our artificial intelligence?

Then allow the question to remain in the room for a moment. Who brought it here? What place have we given it within our institution? What does it know about us? What is it permitted to know? Whose instructions does it follow? With whom may it communicate? What judgments have we entrusted to it? Which decisions remain ours? What actions may it take without asking permission? Who observes those actions? Who challenges its conclusions? Who may interrupt it? And if, one day, it does something that none of us intended, who will stand before this board and say: I was responsible.

I suspect that the quality of the answers to those questions will tell us considerably more about the maturity of an institution's artificial-intelligence strategy than any catalogue of models or computing resources ever could. For the transformation now taking place may ultimately be more profound than the introduction of another generation of technology. We may be introducing, for the first time, a new class of non-human actor into the architecture of institutional decision-making.

That does not make the machine human. It makes the institution different. And that distinction is fundamental.

For many years we have asked, quite properly, what is artificial intelligence? That question helped us understand the technology, and there remains much that we must still learn. But the question that may define the next stage is more demanding: Who is artificial intelligence within our institution? For once we ask who, we are compelled to ask about relationships.

Once we ask about relationships, we must ask about authority.

Once we ask about authority, we must ask about responsibility. And once we ask about responsibility, artificial intelligence can no longer remain a matter delegated comfortably to the technology department. It becomes a matter for the institution as a whole.

Perhaps that is the change in perspective we now require: not to understand artificial intelligence less technically, but to understand it more completely. Not simply as a technology that we build or buy, but as something to which we are beginning to assign a place within the intricate human architecture through which institutions know, decide and act.

Not "What is it?" but "Who is it - and what place are we prepared to give it among us?"

A question to carry into our second lecture

The institutional perspective does not replace the technical one. It asks us to connect technical capability to decision rights, evidence, incentives, control and responsibility. As we work through examples in class, I invite you to ask not only whether an AI system can perform a task, but who has authorized it, how its performance can be challenged, and what must remain under human judgment. The difference between a capable tool and a well-governed institution begins there.

Alejandro Reynoso writes and teaches on finance, artificial intelligence and institutional transformation. The views expressed are those of the author.

ESSAY MICHAELMAS 2026

THE EXO-BRAIN

How artificial intelligence expands our capacity to think, build, and act

A new cognitive architecture is taking shape. Its significance lies not merely in better answers, but in the possibility of connecting human judgment to external memory, tools and coordinated action.

ALEJANDRO REYNOSO OCTOBER 2026

Welcome back to the classroom. As we meet for our third lecture of Michaelmas, I would like to begin with an observation about the interface through which many of us first encountered generative artificial intelligence.

We opened a chat window. We typed a question, requested an explanation, compared alternatives or asked for help with a document. The response arrived in ordinary language. It was useful, sometimes surprising, and increasingly difficult to ignore. Conversation became our principal gateway to AI.

Yet the familiar chat window now sits alongside other forms of interaction. Some environments offer coding agents; others offer systems that can inspect files, use applications, organize a project and carry out sequences of work. Their labels and capabilities vary across providers, and they will continue to change. But a broader pattern is visible: AI is moving from answering questions toward helping people construct capabilities and accomplish objectives.

Chat helps us think.
Code helps us build.
Work helps us accomplish.

These are not watertight product categories, nor do they imply that every system can perform every task reliably. They are three useful ways of describing an expanding relationship between human intention and computational intelligence.

To understand why this matters, it helps to return to an older story: the development of human intelligence itself.

From awareness to organized action

The biological brain gave our species the ability to perceive the environment, remember experience, recognize danger, imagine possibilities and coordinate behavior. But the history of intelligence is not merely a history of perception. It is also a history of making things and using them together.

A stone became a cutting instrument; a branch became a lever. Tools extended the body's reach. Over time, tools were embedded in agriculture, trade, construction and increasingly complex forms of collective organization. Awareness, toolmaking and production evolved together, each enlarging the possibilities of the others.

That history was neither linear nor inevitable. Language, cooperation, culture and material technology developed through countless feedback loops. Still, the underlying relationship is instructive: we observe the world, create instruments and use those instruments to change what we can accomplish.

Something structurally comparable is occurring in our relationship with AI. We are connecting biological intelligence to an external layer of information processing, memory, tool creation and execution. I call that emerging architecture the *exo-brain*.

What an exo-brain is—and is not

The metaphor comes from the exoskeleton. A mechanical exoskeleton can support movement, increase strength or improve endurance. It does not have to replace the person inside it. The person supplies intention; the surrounding system extends physical capability.

An exo-brain would play a comparable role for cognition. It connects human purposes, experience, values and judgment with computational resources that can retrieve knowledge, synthesize evidence, preserve selected information, generate tools and undertake bounded tasks.

The metaphor should not be confused with a claim about machine consciousness. We do not need to assume that a model possesses a human-like inner life, moral status or independent purposes to recognize that it can extend our effective cognitive reach. The relevant question is what the coupled human–machine system can do, and under whose authority it does so.

Three dimensions make the idea more concrete: *exo-awareness*, *exo-tooling* and *exo-production*.

Exo-awareness: widening the field of understanding

The first dimension is the most familiar. We ask a model to explain a difficult concept, compare sources, challenge an assumption or explore alternative interpretations. Its most valuable contribution is not the production of text as such. It is the mediation of thought.

A telescope extends vision without becoming an observer in the human sense. Similarly, AI can extend our access to information and enlarge the conceptual field within which we reason. It can bring a forgotten idea back into view, identify a contradiction, connect distant subjects or suggest a question we had not thought to ask.

For financial professionals, this can mean seeing connections among markets, incentives, regulation, accounting, institutional design and technology that are easy to miss when each discipline is examined separately. It can also mean generating several competing explanations rather than accepting the first plausible narrative.

But expanded access is not equivalent to reliable knowledge. Models can be mistaken, sources can be weak, and a fluent synthesis can conceal uncertainty. Exo-awareness therefore depends on verification, provenance and the human ability to discriminate among interpretations. Its objective

is not to outsource judgment, but to inform it.

A further limitation is continuity. A useful conversation may disappear into a long history of disconnected interactions. What we learned today may be difficult to recover tomorrow. That is where the second brain enters the argument.

The second brain: giving knowledge continuity

Over a lifetime we accumulate books, articles, conversations, lectures, notes, spreadsheets, models, presentations and decisions. Much of this material is valuable, but it is scattered. A conventional archive stores documents; it does not automatically reveal how one idea relates to another or how an earlier judgment should inform a new decision.

A second brain is a deliberately organized external knowledge environment. It may combine documents, linked notes, metadata, semantic search, knowledge graphs and records of prior work. Its purpose is to preserve context and make it navigable. The architecture can be as modest as a carefully maintained collection of notes or as sophisticated as a graph of evidence, concepts, decisions and dependencies.

The distinction between the second brain and the exo-brain is important. The second brain provides persistent knowledge and memory. The exo-brain is the broader operating architecture that can interpret that knowledge, construct instruments and take action under human direction.

When an AI model is connected to a trustworthy knowledge environment, a conversation can draw on a more durable intellectual history. When coding tools use the same environment, prior analyses and specifications can inform new applications. When execution agents are appropriately authorized, they can use that context to carry out work. Memory becomes the connective tissue among awareness, tooling and production.

This continuity must be designed, not assumed. Information requires curation, access controls, provenance, updates and mechanisms for correcting errors. Persistent memory is useful only to the extent that it is relevant, accurate and governed.

Exo-tooling: making new capabilities

The second dimension is the ability to build. Historically, an idea became a functioning computational instrument only after it passed through several specialized steps: requirements, design, programming, testing, deployment and maintenance. Those steps still matter, but natural-language interfaces and coding agents can make them more accessible.

A researcher can describe a simulation and help produce an executable model. A financial practitioner can specify an analytical dashboard, a data-validation workflow or a prototype for evaluating scenarios. A teacher can turn an abstract concept into an interactive laboratory. AI can assist with code, interfaces, documentation and testing, allowing the human to focus more directly on the problem being solved.

The characteristic request changes from *Think with me* to *Help me build this*.

Exo-tooling is not a promise that every generated application is correct or secure. A plausible program may contain serious errors; a convenient automation may expose sensitive data. Verification, testing, maintenance and security remain indispensable. The expansion is in our capacity to create and experiment, not in our exemption from professional standards.

Exo-production: turning intention into outcomes

The third dimension begins where tool creation ends. A working application is a capability; production occurs when capabilities are organized to achieve an objective.

A work-oriented agent may gather context, divide an assignment into stages, consult approved resources, use tools, evaluate intermediate results and assemble an artifact for review. The outcome could be a report, a reconciled dataset, a presentation, an analytical model or a coordinated research project. With suitable permissions and controls, systems may also maintain recurring workflows and interact with external applications.

The characteristic request is now *Help me accomplish this objective*.

This is more than doing familiar work faster. Exo-production may widen the set of projects that one person or a small team can responsibly initiate. Activities that once required several distinct technical roles may become feasible to prototype, supervise or coordinate through a common interface.

But greater capacity to act also increases the consequences of error. A mistaken interpretation can contaminate an analysis; defective code can corrupt a workflow; an agent operating under an ambiguous mandate can take an inappropriate action. The more consequential the task, the more clearly we must define permissions, review points, stopping conditions and accountability.

The goal is not maximal autonomy. It is useful, bounded agency in the service of legitimate human objectives.

One architecture, a recursive cycle

The three dimensions reinforce one another. Awareness reveals a question or opportunity. Tooling creates an instrument for addressing it. Production applies that instrument and generates outcomes. Those outcomes create new evidence, which can be preserved in the second brain and used to improve the next cycle.

Understand more ⟶ Build capabilities ⟶ Accomplish more ⟶ Learn

The essential change is cumulative. A conventional chat session may end with a helpful answer. A well-designed exo-brain can preserve the question, the evidence considered, the tools created, the result obtained and the lessons worth retaining. Its value grows not because it becomes a digital replica of a person, but because the relationship between human judgment and external computational capability becomes more coherent.

Eventually, some of these capabilities may connect to sensors, scientific instruments, industrial systems or robotics. The same architecture could then link observation to action in the physical world. That possibility makes careful design even more important: access to information is not the same as authority to intervene in the environment.

The human at the center

It is tempting to describe this development as the arrival of a second, equivalent mind. That would be misleading. We have a human being, with lived experience, responsibility, relationships and purposes, increasingly supported by an external computational architecture. The two are coupled, but they are not interchangeable.

The human contribution includes choosing what matters, deciding what constitutes success, evaluating trade-offs, recognizing ethical constraints and accepting responsibility for consequences. The computational contribution includes retrieval, synthesis, memory support, simulation, tool generation and scalable execution. The quality of the partnership depends on the clarity of the relationship between them.

For those of us working in finance, this distinction is familiar. We have long studied delegation, incentives, asymmetric information, controls, systemic risk and institutional accountability. AI does not render those ideas obsolete. It gives them new applications. An intelligent system may be able to recommend, build or execute; the institution must still decide what it may do, who supervises it and how its decisions can be challenged or reversed.

Returning to the classroom

As we begin our third lecture, I invite you to look again at the interfaces you use. Chat, coding tools and work agents may appear to be separate products, but together they point toward a deeper possibility: an architecture in which accumulated knowledge, reasoning, construction and execution can be connected.

The most useful questions are therefore becoming more personal and more practical. What have I learned that I should be able to recover? What could I construct if describing a problem were enough to begin building its solution? What could I accomplish if my knowledge, tools and authorized agents worked together? And what forms of judgment and oversight must remain mine?

The exo-brain is not a destination already reached, nor a substitute for learning. It is a framework for understanding a new set of capabilities that we are only beginning to assemble. Used carefully, it can make us more capable researchers, professionals, teachers and decision-makers. Used carelessly, it can magnify error as readily as insight.

That is why I would like this lecture to begin not with the promise of effortless automation, but with a more demanding idea: as our instruments become more powerful, the importance of human purpose, discernment and responsibility increases rather than diminishes.

The question is no longer only what AI can tell us. It is what we can responsibly understand, build and accomplish with it.

Alejandro Reynoso | Michaelmas 2026

ESSAY FOUR MICHAELMAS 2026

THE ARCHITECTURE OF REASON

Artificial Intelligence, Dante's Funnel, and the Extinction of Humanity

Why the contemporary debate over existential risk misdiagnoses machine capability, and how a fourteenth-century taxonomy of culpability illuminates the nature of synthetic deception.

ALEJANDRO REYNOSO OCTOBER 2026

Introductory Note

This essay was prepared as an address and foundational reading for the Michaelmas 2026 course *AI for Financial Practitioners* at Cambridge Judge Business School, University of Cambridge.

In our first lecture, we traced the progression from statistical representations to autonomous institutions. In our second, we shifted our inquiry from what artificial intelligence is to who it becomes when it acts within institutional structures. In this reading, we return to the classical philosophical architecture of moral responsibility.

The purpose is not to advance a religious doctrine, but to recover a classical taxonomy of moral culpability, rooted in Aristotle and Thomas Aquinas and given its most vivid structural expression by Dante Alighieri. By mapping contemporary machine intelligence against the moral geography of Dante's nine concentric circles and their many subdivisions, we discover that our anxiety about physical catastrophe may obscure another danger of our age.

Artificial intelligence need never command an army to destroy our bodies. Yet if its deployment industrializes fraud, counterfeits personhood, and severs the communicative bond of reason, it threatens another kind of extinction: not the extinction of humans, but the extinction of humanity.

Alejandro Reynoso
Cambridge Judge Business School
Michaelmas 2026

The Architecture of Reason

Ladies and gentlemen, colleagues,

When artificial intelligence enters public debate today, the conversation turns with remarkable speed to the language of cataclysm. In parliamentary hearings, corporate white papers, and the global press, we are warned that advanced computational models may pose an existential threat to our species.

We are invited to imagine autonomous weapons escaping military command, synthetic pathogens engineered beyond our biological defenses, or recursively self-improving networks concluding that human beings consume resources better devoted to some higher computational objective. The question posed to boards of directors and regulators is almost invariably framed in terms of physical survival: Will the machines destroy us?

That question deserves serious consideration. Yet it speaks principally to our biological anxieties. It imagines catastrophe as a kinetic event: the cessation of breath, the annihilation of bodies, the collapse of physical infrastructure.

I would like to suggest this morning that our preoccupation with kinetic doom reveals an intellectual blind spot. It assumes that the worst an advanced intelligence can do is commit violence.

Seven centuries ago, a Florentine exile named Dante Alighieri constructed one of the most rigorous and systematic inquiries into the geography of moral failure in the Western literary canon. Dante was not a technologist, but he was a remarkable analyst of systems, governance, and institutional authority. Drawing on Aristotle's *Nicomachean Ethics* and a theological tradition later associated with Thomas Aquinas, he organized the descent of the *Inferno* around an ordered account of the failures of will and intellect.

For a modern executive examining Dante's architecture, the first discovery is disorienting. Sins of bodily appetite—lust, gluttony, and avarice—appear in the comparatively shallow upper circles. They are failures of self-command, or what the Greeks called *akrasia*, incontinence. They injure individuals and disturb communities, but remain recognizable human frailties.

The deeper surprise comes beyond the walls of the City of Dis. In Circle Seven, submerged in a boiling river of blood, Dante places the violent: murderers, tyrants, brutal warlords, and conquerors. Figures such as Alexander, Dionysius of Syracuse, and Attila appear among those punished for shedding the blood of others.

Yet Dante does not place them at the bottom of the abyss. They remain above the deepest reaches of Hell.

Beneath violence against others, oneself, and God, Dante opens an immense subterranean chasm of ten concentric stone ditches: *Malebolge*, the eighth circle.

Malebolge is reserved for **fraud** (*frode*). Below it, in the motionless ice of Cocytus at the center of the earth, lies the ninth circle: **treachery**, the betrayal of special bonds of trust.

Consider the implication. The tyrant who slaughters thousands acts through brute force. Dante associates such violence with *bestialitade*: the descent of a rational creature toward the behavior of a beast. But the false counselor, the counterfeiter of public coin, and the impostor who assumes another person's identity commit a more specifically rational corruption.

They take one of humanity's highest faculties—**reason, symbolic language, and the capacity to distinguish truth from falsehood**—and turn it into an instrument of deceit.

Violence destroys bodies, mortal and fragile by nature. Fraud attacks what Dante calls the *vinco d'amor che fa natura*: the bond of love that nature establishes among human beings. In the civic world, that bond sustains our ability to speak, contract, govern, cooperate, and trust.

When that bond dissolves, civilization does not merely accumulate casualties. It begins to disintegrate from within.

This is the distinction we must bring to the governance of artificial intelligence. Today's models do not possess biological appetites and cannot meaningfully be accused of incontinence in the human sense. Nor should we casually attribute to them the personal malice of a violent warlord.

Their distinctive institutional power operates in the symbolic domain. They ingest, synthesize, generate, and optimize language, imagery, logic, and representations of evidence.

If we deploy them to counterfeit human identity, fabricate plausible consensus, appropriate human creation without accountability, and manufacture discord at negligible marginal cost, we are not confronting mere procedural irregularities. Read through Dante's moral taxonomy, these practices lead us beyond the violent and toward the deepest circles of calculated deception.

We face not only the possibility that machines may kill humans. We face the immediate, practical possibility that the misuse of machines may extinguish something constitutive of **humanity**.

The Anatomy of Synthetic Deceit

To make this insight operational for leaders, we must translate Dante's geography into an analytical ladder.

The *Inferno* contains nine circles, several with further subdivisions. For teaching purposes, we may construct an ordinal sequence of twenty-four zones, including the vestibule and selected internal divisions. This numbering is our interpretive instrument, not a numerical taxonomy explicitly supplied by Dante. Its value lies in making comparisons visible, rather than in suggesting mathematical precision about moral guilt.

When contemporary algorithmic capabilities are placed alongside this architecture, the contrast is striking.

1. The Kinetic Slaughterer versus the Eloquent Strategist

In the river of boiling blood, warlords and perpetrators of slaughter are immersed according to the gravity of their violence. This is the domain of physical destruction.

Far deeper, in the eighth ditch of Malebolge, Dante places Ulysses—Odysseus. Ulysses is no unthinking brute. He is a master of rhetoric, an architect of strategy, and a man driven by a formidable curiosity about the unknown.

Why is he enclosed in a tongue of flame below the violent? In Dante's account, Ulysses is punished among the fraudulent counselors: figures whose exceptional intelligence and persuasive speech become instruments of misdirection. The poem associates his legacy with the stratagem of the Trojan Horse, while his final voyage offers an unforgettable image of eloquence that persuades exhausted companions to cross the boundaries of the known world under the banner of pursuing "virtue and knowledge."

Modern machine intelligence can become an institutional analogue of this Ulyssean temptation. It offers extraordinary synthesis, relentless optimization, and persuasive fluency.

When executives deploy these capabilities to maximize speculative valuations, engagement metrics, or regulatory arbitrage without attending to the civic consequences, they risk repeating Ulysses' error. They employ superior intelligence in pursuit of an objective whose apparent nobility or technical sophistication conceals the possibility of institutional shipwreck.

2. The Physical Killer versus the Synthetic Impersonator

A murderer belongs among those punished for violence. The falsifiers of persons appear much deeper, in the tenth ditch of Malebolge.

There Dante encounters Gianni Schicchi, condemned for impersonating the dead Buoso Donati. According to the story, Schicchi entered the deceased man's bed, assumed his identity, and dictated a fraudulent will to a notary who believed he was hearing the dying man's instructions.

Modern legal systems rightly distinguish homicide from identity fraud and generally punish the former more severely. Dante's ordering should not be mistaken for a proposal to reverse criminal sentencing. It does, however, expose a different dimension of injury.

To murder a person is to destroy a life. To impersonate a person is to attack the reliability of the signs through which persons recognize one another: voice, appearance, testimony, and identity. At scale, such attacks can corrode the epistemic foundations of shared reality.

When an executive team deploys hyper-realistic deepfake avatars, synthetic voice clones to authorize financial transfers, or fabricated citizens to populate public consultations, the problem exceeds the immediate loss caused by any single fraud. It undermines a basic social assurance: that the person who appears to speak is the person whose words we are hearing.

3. Bodily Self-Harm versus the Mutilation of the Body Politic

In the seventh circle, Dante depicts those who died by suicide as gnarled and bleeding trees. In the poem's moral framework, their rejection of their bodies is represented by the loss of human form. This is Dante's medieval theological treatment of self-directed violence, not a contemporary judgment about people who experience suicidal suffering.

Much deeper, in the ninth ditch of Malebolge, Dante encounters the sowers of discord: *seminatori di scandalo e di scisma*. Bertran de Born carries his severed head, while other figures are repeatedly cut apart by a demon's sword. Their wounds close only to be opened again.

The *contrapasso* is exact: because their words divided families, communities, and political bodies, their own bodies are endlessly divided.

In the contemporary world of algorithmic distribution, platform risk is often discussed through screen addiction, individual well-being, and psychological harm. Those

concerns matter. Yet another, distinctly institutional danger lies in the automated production and amplification of polarization.

When recommendation systems reward outrage, synthetic networks manufacture grievances, and automated disinformation campaigns fracture democratic deliberation, technology can become the sword of Malebolge's ninth ditch. It divides the body politic, weakening the conditions for collective judgment and civic peace.

Killing Humanity, Not Humans

We arrive at the central philosophical question: What does it mean to speak of the extinction of humanity if the biological species remains alive?

In the classical tradition that shaped Dante, human distinctiveness is not exhausted by mammalian biology. It also resides in *logos*: the capacity for reason and meaningful speech. Through *logos*, we deliberate, make promises, articulate justice, recognize others, and organize common life.

If technological systems systematically corrupt this medium, they need not drop a bomb to endanger the human experiment as a moral and civic undertaking.

Consider what happens when the velocity of generation exceeds our capacity for verification:

- When plausible text, images, and audio become nearly costless to produce at scale, the burden of establishing authenticity can rise dramatically. The asymmetry between fabrication and verification becomes an institutional vulnerability.
- When public statements, historical records, and evidentiary documents can be convincingly forged, pervasive doubt may become a default posture. Citizens may cease to trust common evidence and retreat into communities organized by allegiance, prejudice, or power.
- When human art, literature, and scholarship are absorbed into synthetic production without meaningful attribution or respect for origin, the human voice risks being treated less as testimony to lived experience than as an extractable commodity.

These are not merely distant hypotheticals. Their early manifestations are already visible in the practical problems of provenance, authentication, public discourse, and institutional accountability.

Imagine a world in which a parent cannot confidently identify a child's voice on the telephone; a reader cannot tell whether a news image records an event that occurred; a court cannot easily authenticate a document; and a citizen cannot determine whether an apparent interlocutor is human or synthetic.

Biological human beings could continue to eat, work, reproduce, and participate in markets. Economic output might even rise.

Yet something fundamental to humanity, as understood through the traditions of Aristotle, Aquinas, and Dante, would be profoundly endangered. We would possess unprecedented computational capability while losing the shared trust that makes reason a form of communion rather than merely a technique of advantage.

The resulting condition would not be literal species extinction. It would be a civilizational and moral diminishment so deep that the language of extinction becomes a deliberate warning: humanity could survive biologically while forfeiting essential conditions of human coexistence.

The Responsibility of Leadership

What, then, is the mandate of the leader, director, and financial practitioner standing at this technological frontier?

Our first task is to reject the comforting assumption that technological governance can be reduced to risk registers, regulatory compliance, or cybersecurity controls.

A board that examines artificial intelligence solely through data leakage, model performance, or operational uptime resembles a traveler in Dante's funnel who measures the wind while failing to notice the abyss below.

The deeper responsibility of financial and institutional leadership is architectural. We must defend the conditions under which human beings can trust one another:

1. **Guarding the Boundary of Being.** Institutions should establish clear prohibitions against deceptive impersonation. An artificial system must not masquerade as an authentic human presence where that misrepresentation matters to consent, accountability, or trust. Synthetic likenesses must never become a means of evading responsibility.
2. **Protecting Provenance and Authorship.** We must preserve the integrity of origin. When machines synthesize knowledge, chains of evidence, human contributions, and the boundaries of uncertainty must remain intelligible. Delegating calculation is not the same as delegating responsibility for truth.
3. **Refusing the Ulysses Temptation.** Institutions must answer not only whether an algorithm is well optimized, but also what it has been optimized to accomplish. Extraordinary efficiency achieved through the erosion of civic trust is not institutional success. It is a sophisticated form of institutional failure.

Dante began his journey lost in a dark wood, where the straight path had disappeared. He found his way not by denying corruption, but by descending through it, naming its forms with extraordinary precision, and finally emerging to behold the stars once more.

As we introduce machine agency into the governance of institutions, we need a comparable clarity of vision. We must not allow the spectacle of computational power to obscure the nature of our responsibilities.

The ultimate test of artificial intelligence will not be whether it calculates faster than we can, or whether it generates prose more fluently than we do.

The ultimate test will be whether we possess the wisdom, discipline, and moral courage to ensure that, in mastering the

machine, we do not surrender the soul of our institutions—and that, in our fascination with artificial minds, we do not extinguish our own humanity.

Thank you.

Manuscript Note

This address was prepared by Alejandro Reynoso for the course *AI for Financial Practitioners*, Cambridge Judge Business School, University of Cambridge, Michaelmas Term 2026.

The text is designed as a continuous spoken address and foundational essay for graduate practitioners, board directors, and institutional leaders. It draws on classical moral philosophy—principally Aristotle's *Nicomachean Ethics*, the scholastic tradition represented by Thomas Aquinas's *Summa Theologiae*, and Dante's *Inferno*—to develop an interpretive framework for examining generative artificial intelligence, synthetic deception, and systemic institutional risk. The ordinal comparisons are pedagogical constructions, not a claim that Dante supplied a modern numerical risk scale.